

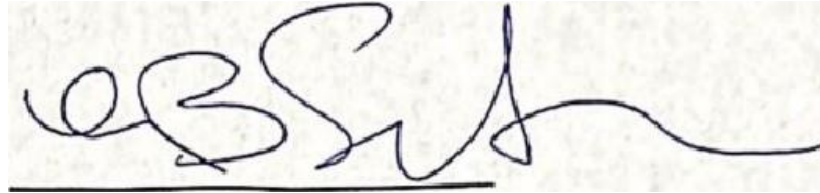
EFFECTS OF INFANT VOCALIZATIONS ON MOMENTS OF TRIADIC ATTENTION

Austin Himebaugh

Submitted to the faculty of the Department of Psychological and Brain Sciences
in partial fulfillment of the requirements for the
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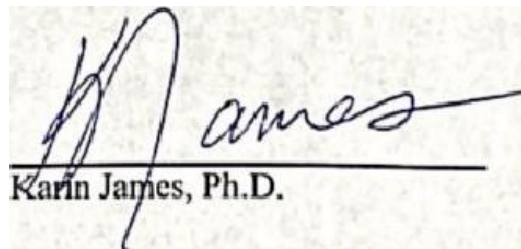
Accepted by the Psychological and Brain Sciences Faculty, Indiana University, in partial fulfillment of the requirements for the Honors degree in Psychology.

A handwritten signature in dark ink, appearing to be 'Linda Smith', written over a horizontal line.

Linda Smith, Ph.D. (Chair)

A handwritten signature in dark ink, appearing to be 'Sara Benham', written over a horizontal line.

Sara Benham, Ph.D.

A handwritten signature in dark ink, appearing to be 'Karin James', written over a horizontal line.

Karin James, Ph.D.

April, 2024

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This thesis is dedicated to all my family members, my mother Nikole, father Bryce, brother Blake, and sister Isabelle. Thank you for believing in me. I want to give a special thanks to Dr. Linda Smith and Dr. Sara Benham for their help guiding me through this process, I could never have done it without them. Also, a big thank you to everyone in the Cognitive Development Lab. Lastly, thank you to Dr. Karin James for serving on my defense committee, as well as Dr. Dale Sengelaub for his wise input and weekly meetings.

Effects of Infant Vocalizations on Moments of Triadic Attention

Abstract

Triadic attention – a form of joint attention involving shared concentration on an object between an infant and secondary person plays a critical role in advancing the foundations of infant language development, particularly object naming. Infant vocalizations are known to elicit object naming by parents. However, whether infant vocalizations promote the occurrence of triadic attention is an open question that has not been widely studied. To characterize the key components of triadic attention, I analyzed video recordings from a camera worn by infants in the home, focusing on an age period (7-11 months) during which triadic attention first and fragilely begins. I coded 5-second intervals of video for the presence of adult hands, adult faces, and objects handled by the child as indices of shared concentration on an object. I also recorded the presence of child vocalizations in the 5-second interval immediately prior to the triadic attention event. I hypothesized that triadic attention sequences initiated by an infant vocalization would be more frequent than triadic attention events with no vocalizations, so infants would vocalize more during moments just prior to triadic attention sequences. However, the results show that triadic attention events occurred more with no infant vocalizations immediately before than with infant vocalizations immediately before. The findings from this study speak to the inconsistent nature of triadic attention moments before the one-year mark, as well as relates a specific age group's attention moments to their vocalization frequency and timing.

Introduction

Attention is an essential component to social interactions. Quality interactions depend on attending to the social partners words, faces, gestures, and the physical context. Young infants engage socially in face-to-face play with a mature partner (Lavelli & Fogel, 2013). At 3 months infants are sharing mutual gaze with their caregiver, at 6 months they are manipulating their caregiver's attention (Kaplan & Hafner, 2006), and by 9 months developing and exhibiting what is known as triadic attention - shared attention with a partner with respect to an object. This form of attention –another person and an object - is the central to play, to joint action, and to learning words (Baldwin, 1995).

Triadic attention is a form of joint attention with three components: an object, a child or caregiver initiating gaze towards an object, and the opposite side of the partnership following the other's gaze and looking at the object (Yu & Smith, 2017). Although original approaches to triadic attention focused narrowly on gaze (Farroni et al., 2004; Vecera & Johnson, 1995), more recent research has broadened measures to multimodal ways of attending including touching and handling objects (Yu & Smith, 2013). As both infant gaze and object handling impact triadic attention, both were tracked in this study.

Triadic attention allows for social coordination that elicits language learning. When children explore objects, parents tend to label them. Joint attention is crucial to this process, as children who are in this attentive state are more receptive to these object labels and acquire language faster (Baldwin, 1995). In its simplest form, triadic attention occurs when Person A is paying attention to Person B, Person B is paying attention to Person A, and both A and B are paying attention to an object simultaneously (Striano & Stahl, 2005). Triadic attention is integral to the

infant learning experience, as it is involved in deciphering social cues such as emotion tracking, eye contact, body movement (Striano et al., 2006) and gaze following (Deak et al., 2014).

Triadic Attention and Object Focus

The object - the entity that connects participants of triadic attention – is the centerpiece to triadic attention and the component that separates early face to face social interactions from the social context of learning about the world. During moments of triadic attention, infants tend to lock in on the object of interest, and neglect most other things in sight, often including their partner. Yu and Smith (2017) equipped 11-to-24-month-olds with head-mounted eye trackers and had them and their parents play with different toys in the lab. They found that infants rarely looked at their parents during the play session and spent most of the time looking at the object instead. Parents, in contrast, were much more likely to look at their infant's face and spent only a fraction of the time attending to the object of interest.

Yu and Smith's (2017) study provides evidence of infants achieving triadic attention through manipulation of objects, as well as positing that gaze following is not always the main, or earliest, factor in maintaining triadic attention. A question that this study does not address, however, is whether gaze following has any sort of impact on moments of triadic attention.

Theories on how gaze and hand-following are related, as well as their role in promoting triadic attention, have evolved over time. It is now widely accepted that infants tend to look at objects, while parents tend to look at faces. But there are times when infants do look at faces, and infants tend to do something interesting (perhaps wanting to share with parents) right after looking. In a study done by Deak et al. (2014), they placed cameras on the heads of infants during a free play period of 7 to 9 minutes and an object play period of 7 to 10 minutes. The group found that when an infant looked at their caregiver while the caregiver was looking at their

own hands, the infant's next eye movement was towards the parent object-handling, a clear indicator of gaze following to the parent's hands. Other evidence shows that gaze following increases with age (Yu & Smith, 2017).

Adult Verbal and Non-Verbal Cues

Along with object handling and gaze-following to the object of interest, smiling and parent vocalizing towards infants have been found to influence moments of triadic attention. In a study by Striano & Stahl (2005), young infants, 3 to 9-month-olds interacted with a female stranger and a toy across a 5-minute interaction. Each infant received four separate conditions lasting 30 seconds. These were: "joint attention" phase; where the adult would hold their gaze at the object of interest, look at the face for two seconds, and smile and talk; "alternate only" phase, where the adult did the exact same thing as the "joint attention" phase, but did not smile or talk; the "affect only" phase, which was the same as the "join attention" phase, but the adult would break eye contact with the infant for one second before looking back at the object in order for the adult's smile and talking to not be interpreted towards the object; and the "ignore" phase, where the adult only looked at the object, and did not vocalize or smile.

The researchers found that infants gazed significantly more at the adult during the "joint attention" condition, which supports the idea that adult non-verbal (smiling) and verbal (maternal partner vocalizations) enhance moments of triadic attention. Previous research has not examined infant vocalizations in moments of triadic attention.

Infant Vocalizations

Past research has shown that vocalizations put infants in a state that is conducive to learning. Goldstein et al. (2010) counted the number of object-directed vocalizations infants used towards each played-with object. Then, they completed a preferential looking task where they compared

images of objects they that had occurred with infant vocalization. The shown images were degraded (blurry) versions of the played with objects, which should limit recognition. Infants performed better, recognizing degraded images, when they had produced higher numbers of object-directed vocalizations.

In a second experiment of the same study, the researchers presented the infants with another preferential learning task. The experimenters either labeled an object in response to an object-directed vocalization by the infant, or they labeled the object in response to an infant look without vocalization. When subsequently tested with degraded images of the object, infants sustained visual attention to the images when they had vocalized during the labeling phase, providing evidence that infant vocalizations are correlated with moments of increased attention and learning.

These findings establish a link between infant vocalizations and triadic attention, but it does not reveal the causal pathway. Are object-directed vocalizations occurring because of an increase in focused attention, or do vocalizations increase arousal and put infants in states of heightened attention? Infants better learn features of objects that elicit the most object-directed vocalizations (Goldstein et al., 2010). Therefore, vocalizations may assist in locking infants into moments of triadic attention. If this is the case, there could be important implications for triadic attention initiation, object learning, and vocabulary size. A child's vocalization would signal the ideal moment for parents to label an object (Goldstein & Schwade, 2010), as the vocalization is an indicator of the child's willingness to learn that object (Goldstein et al., 2010). If the parent is more likely to label an object and a child is more likely to be prepared to receive that object label, early word learning and vocabulary size may be enhanced (Goldstein & Schwade, 2010;

Goldstein et al., 2010). Finally, the Goldstein et al. studies occur in the laboratory in experimenter-designed contexts that may increase arousal and thus infant vocalization.

In this study, I sought out to investigate whether infant vocalizations elicit moments of triadic attention in natural at home and daily-life interactions. To do this, I coded at-home videos of 7 to 11-month-olds from the Homeview Project dataset, (<https://cogdev.siteshost.iu.edu/homeview.html#Welcome>) a data set of day-long head camera recordings collected for the purpose of studying visual (not specifically social) learning of infants in the home. In the sampled data from the larger corpus, I examined contexts of social play.

Age Range and Relationship to Infant Vocalizations

This infant age range (7-11 months) was ideal because 9 months is an extremely important stage of vocal development for infants. At this age, infants are babbling and producing more speech-like sequences than ever, which are more advanced and predictive of language development (Nathani et al., 2006). Specifically, infants in the 9–12-month range are beginning to produce speech-like sequences as they transition to their first words. So, the data collected in this study occurs right in the middle of a vocalization transition that is extremely formative and lays the groundwork for later language development (Vihman et al., 1985). With more speech-like utterances available at this age, it is the ideal time to examine their relationship to the parent-infant attention relationship.

For each subject, I noted each moment of triadic attention, each infant vocalization, and counted how many times an infant vocalization came 5-seconds or less before a moment of triadic attention.

If vocalizations play a role in the daily-life experiences that support the development of triadic attention, then infant vocalizations should occur in those contexts. Moreover, if infant vocalizations elicit parent responses and create joint attention, then, consistent with a Goldstein et al. (2010), triadic attention sequences with an infant vocalization would be more frequent than triadic attention events with no vocalizations. Observing this would provide evidence showing critical infant vocalizations may be prior to the onset of triadic attention.

Methods

Participants

This study used 3 participants from the Bloomington, Indiana area. Participant ages were 29 weeks, 39 weeks, and 43 weeks, with an average age being 37 weeks old (7-11 months).

Materials

This study involves three main pieces of equipment: the Looxcie 2 wearable video camera, a cloth hat, and the ELAN (EUDICO Linguistic Annotator, <https://archive.mpi.nl/tla/elan>) audio/video annotation software. Cameras were attached to hats, and the infants were unaware of the camera while wearing the hat. ELAN software was used to code various actions/events that the infant might encounter on camera.

Procedure

Over the previous years, the Cognitive Development Lab has collected videos of households from the infant's point of view. Upon agreement, a hat/camera combination is dropped off to the family and instructed on the basic details (how to turn it on, making sure it stays on the infant's head, etc.). Once parents learn the rules and basic functions, they are instructed to place it on their infant's head and leave it for the hours that they want to record video. Over that time, video

and audio data is collected to capture on camera what an infant sees and hears while going about their daily life.

Splitting Data into Five Second Bins

Once data were collected, the data were coded. The coding process began by obtaining the footage from the cameras and exporting it to ELAN. In ELAN, videos were split into 5-second bins, so we were able to navigate in 5-second intervals across the data. To do this, the raw data file was imported to ELAN. Once the video was in the ELAN software, the Skip-to-End button was clicked, which took the icon that controlled where you were in the video to the very end of the video. Next, in the tab section at the top of the screen, the Tier tab was selected, and then within that tab the “Create Regular Annotation” section was selected. Once that was selected, a pop-up window would come onto the screen. In this window, you selected the Bin tier at the top of the screen, and also had the opportunity to set the Start Time, End Time, Duration, and Annotation Size. The Start Time for the video would always be 0:00:00.000 because we always wanted to start coding at the very beginning of the video. Similarly, the End Time and Duration were both set to the video’s total length. If the video was 5 minutes, 10 seconds, and 4 milliseconds long, in ELAN these two categories would be set to 00:05:10.004. Adjusting the Annotation Size was the method to split the video into bins, where you could code the different events occurring in each bin separately. For this study 5-second bins were to be used, so the Annotation Size was to be 00:00:05.000. After these steps were completed, the video would be split into 5-second bins that can have different event, location, and vocalization events occurring simultaneously.

Interval Labels

Event Coding:

When ELAN was introduced to the lab, lab members made specific labels that coders can label each bin with. They are split into different categories and subcategories, with the first being under the “Event” type. To code for each Event, we clicked on the word “Event”, selected “Yes” from a drop-down menu, clicked on the event subcategory or categories occurring within the 5-second interval, and selected “Yes” again. If it was unclear what the baby was doing in that five-second interval and there was no clear event occurring, that bin was coded as a “Watching” event. To aid coders and provide specific criteria on where the baby is in each interval, the lab created a coding manual explaining what needs to be in a room and in view to be considered a specific event. Each event, criteria, and examples are listed in Table 1.

The only subcategory that contained its own categories was the Adult in View subcategory. While coding, if the Adult in View subcategory was clicked “Yes”, it brought an option list with choices of “Face”, “Hand”, or “Both”, signaling whether the infant was looking at a parent’s face, hands, or both face and hands during the five-second bin.

Vocalization Coding:

The second coding pass completed was searching for infant vocalizations. For each five-second interval, we tracked whether the infant with the camera on vocalized. The same coding manual was used to determine what to consider as a vocalization. There were only two vocalization events to code, Child Speech and Other Speech. Infant vocalization types that were coded were cries, syllables, babbling, grunts, cooing, words, and singing. These are all considered speech-like vocalizations, as each one of these utterances lies on a developmental timeline where each one will grow closer to full speech as the infant gets older (Oller, 2000). Vocalization types that were not coded were hiccups, burps, laughing, and sneezing. The reason why these four vocalization types were not included was because they are considered non

Event Codes	Elements	Examples
Object Handling	Child who is wearing the camera is grasping an object that is movable/if the child is acting on an object	Banging pots, waving a spoon around, grabbing mom's necklace count as object handling Touching or sitting on a chair is not handling, but grasping the chair and pushing it is Sitting on a bicycle is not handling, but holding the handlebars is
Adult in View Subevents: Face, Hand, Both	Any part of the adult is in the child's view	Adult's hand, shoulder, leg, face – any part of an adult is in view

Table 1. Each event's criteria for coding it for a 5-second bin, as well as examples for each event.

speech-like vocalizations, which are not considered precursors to speech during development (Oller, 2000). The same method for coding each vocalization was used, and if there was a lack of an infant vocalization, then we did not code anything. The different types of vocalizations are listed in Table 2.

Defining an Event in ELAN

An event was defined as any action that occurred within the five-second bin of a video. For example, if an infant or adult was shown in the video handling an object within a five-second bin, that would be considered an “object handling” event. If the infant or parent continued to handle that same object in the next five-second bin, we would have two “object handling” events, which in totality would last ten seconds. Even if the handling lasted less than five seconds, it was still considered an “event” that lasted five seconds.

Defining Triadic Attention in ELAN

After initial coding, we defined triadic attention in ELAN. In this study, three criteria needed to be present in the same 5-second bin to consider it a moment of triadic attention:

1. An adult in view (either their face, hand, or both)
2. An object in view
3. That same object must be on camera either being handled by the parent or infant.

If all three were present, that five-second bin was considered a moment of triadic attention, also known in our study as a triadic attention event.

Data Analysis

Counting Lengths, Proportions, and Events/Minute for Aspects of Triadic Attention

For each subject, the number of events were coded for three groups: Adult Hand or Face in View, Object Handling, and Infant Vocalizations. To obtain the events for Adult Hand or Face in

Vocalizations Codes	Elements	Examples
Child Speech	Child wearing the camera making any vocal sound	<u>What does count:</u> -cries -syllables -babbling -grunts -cooing sounds -words (whether you can understand them or not) -singing <u>What doesn't count as child speech:</u> -hiccups -burps -laughing -sneezing

Table 2. The criteria for a coded infant vocalization, as well as examples for each vocalization type.

View, all events from the Adult in View subcategory that had either a parent hand, face, or both were included. All the events in the Object Handling subcategory were included. For the Infant Vocalization group, all the Child Speech events were used. After the number of events for each new group were found, that number was multiplied by 5 – because each bin is 5 seconds – to obtain the total amount of time with Adult Hand or Face in View, Object Handling, and Infant Vocalizations.

After that, the total time for each event (Adult Hand or Face in View, Object Handling, Infant Vocalizations) was divided by the overall total time of the video to obtain the proportion of each event. To calculate events per minute, the total number of events for each event was added together, and then divided by three (three subjects), to find the average number of events for that subject in that event. Then, the total amount of video time was divided by three (three subjects) to find the average length of a subject's videos. After that, each category's average event number was divided by the average length of a subject's video to get events/minute.

Finding Time Spent in Triadic Attention

Each bin that contained both an Adult Hand or Face in View and Object Handling were counted as Triadic Attention events. Every other bin was considered a Non-Triadic event. The number of bins for each of those groups were counted separately, and also multiplied by 5 separately to obtain the total amount of time for each of those events. To obtain the proportion of time spent in triadic attention compared to time spent outside of triadic attention, Triadic and Non-Triadic event times were divided by the total video length times. Each subject's time in triadic attention was also tracked. This was done by finding all of the bins containing Adult Hand or Face in View and Object Handling for each individual subject. Additionally, the time spent in triadic attention was divided by the subject's video time to find the proportion spent in triadic

attention for each subject. In addition, the minutes spent in triadic attention was divided by the total time (in hours) to obtain the number of minutes per hour of triadic attention for each subject.

Finding Number of Triadic Attention Events Occurring Immediately After Infant Vocalization

The amount of time infants spent in triadic attention was tracked by counting the five-second bins containing Adult Hand or Face in View and Object Handling. After those were counted, that number was split by whether that Triadic Attention bin was preceded by a five-second bin containing an infant vocalization (also know as the Child Speech vocalization event). Those number of bins were both multiplied by 5 to get the total time for each triadic attention group. Both the infant vocalization group and the non-infant-vocalization group times were divided by the total triadic attention time to get the proportion of triadic attention events containing an infant vocalization immediately before, and the triadic attention events not containing an infant vocalization immediately before.

Chi-Square Test of Association

A Chi-Square Test of Association was performed to determine whether there was an association between vocalizations prior to different events and the occurrence of triadic attention events. To perform this, the Jamovi data analysis software was used to create contingency tables and χ^2 data figures (χ^2 value, degrees of freedom, p-value, number of events).

Contingency Table:

Contingency tables were created for each subject, as well as a total contingency table for all subjects combined. The table contained the number of triadic attention events where an infant vocalization occurred immediately before, the number of infant vocalization events where there

was not a moment of triadic attention immediately after, the number of triadic attention events that did not contain an infant vocalization immediately before, and the number of events where a non-triadic event did also did not contain an infant vocalization immediately before. These four groups were obtained by counting every time these co-occurred in ELAN.

Finding Infant Vocalizations Prior to Aspects of Triadic Attention vs. Triadic Attention

The amount of time where an infant vocalized immediately before only looking at a parent's face was compared to the amount of time where an infant vocalized before immediately looking at both a parent's face and hand in the same bin, as well as compared to the amount of time an infant vocalized immediately before a moment of triadic attention. For the "Infant Vocalization Before Looking at Parent Face" group, the number of Child Speech bins that occurred directly prior to an Adult in View bin with only "Face" option were selected. For the "Infant Vocalization Before Face and Hand" group, the number of Child Speech bins that occurred directly prior to an Adult in View bin with only the "Both" option were selected. For the "Infant Vocalization Before Triadic Attention" group, the number of Child Speech bins that occurred directly prior to a bin that contained any Adult in View option and Object Handling were selected. After all bins for each group were counted, they were multiplied by 5 to get the total time for each group. Then each group was divided by the total amount of time infants vocalized across all videos to their respective proportion related to how much each group occurs compared to all of infant vocalizations.

Results

Aspects of Triadic Attention

Figure 1 shows the frequency of infant attention to parent face and/or hand as indices of parent attention and to the object that is the focus of parent attention, object handling, and

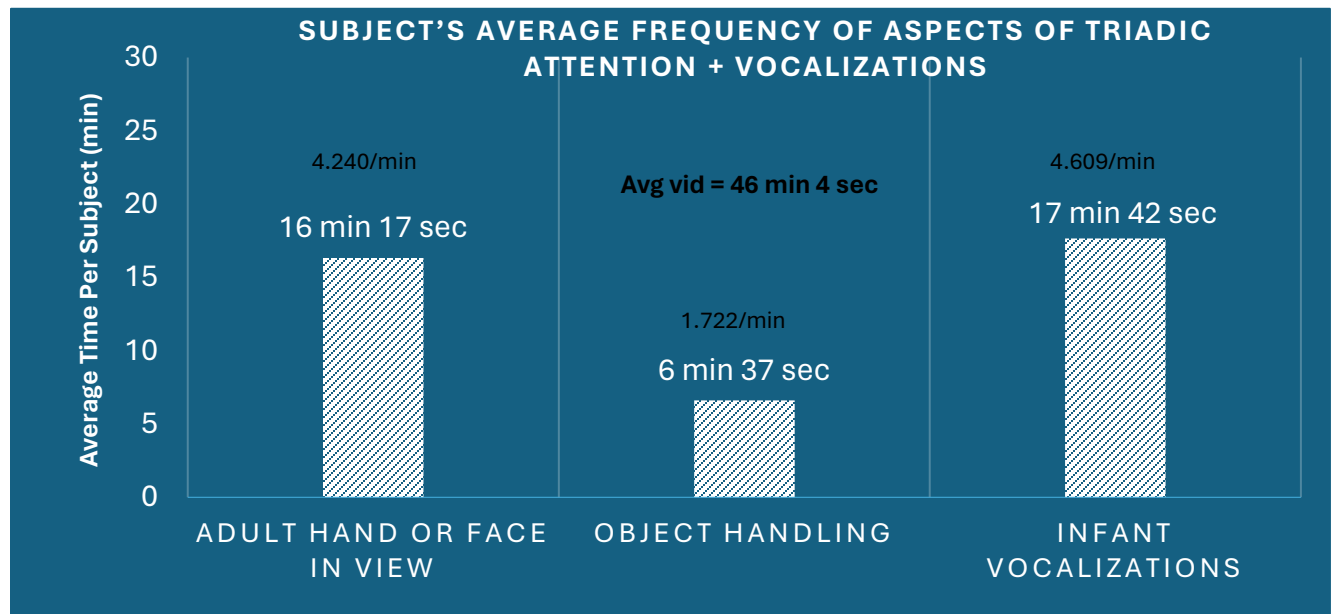


Figure 1. The average subject's time spent looking at both aspects of triadic attention (adult hand/face in view and object handling) and triadic attention throughout a video. The graph also gives that data in instances per minute, as well as lists the average length of a video.

vocalization. Across all videos (2 hours, 18 minutes, and 13 seconds, $SD = 11.10$ or 11 minutes and 6 seconds), infants and adults handled objects for 19 minutes and 50 seconds (1.722 events/minute, 14.349% of total time), an adult's hand or face was in view for 48 minutes and 50 seconds (4.240 events/minute, 35.331% of total time), and the infant vocalized for 53 minutes and 5 seconds (4.609 events/minute, 38.406% of total time). Table 3 lists each infant's total video contributions, time spent with an adult in view, time spent handling objects, and time spent vocalizing.

Referring to Figure 1 and Table 4, infants varied in the measured behaviors. Events per minute, the mean of the three participants divided by the mean of total time of the coded videos provides a normalization of the patterns across subjects and is thus the principal measure.

Time Spent in Triadic Attention

Figure 2 compares the total time infants spent inside triadic attention to the total time they spent outside of it. Across all videos, triadic attention events occurred at a rate of 6.252 minutes/hour. These results suggest that triadic attention occurs regularly at the ages between 29-43 weeks. The results of Figure 3 are also consistent with the literature regarding infant age and triadic attention frequency. For this graph, each subject's time in triadic attention was tracked by finding all the five-second bins containing object handling and an adult hand or face in view. Additionally, the time spent in triadic attention was divided by the total time to find the proportion spent in triadic attention for each subject. Also, the number of five-second bins were counted, and then divided by the total time to get events per minute. This process revealed that Subject 26772 (29 weeks, video length 57 minutes and 8 seconds) had the shortest time of triadic attention at 1 minute and 15 seconds (0.022 of total time), Subject 27959 (39 weeks, video length 30 minutes and 49 seconds) produced 4 minutes and 50 seconds of triadic attention (0.157 of

	Subject 26772	Subject 27959	Subject 26445	Total
Age	29 weeks	39 weeks	43 weeks	Avg = 37 weeks
Video Contributed	57 min 08 sec	30 min 49 sec	50 min 14 sec	2 hr 18 min 13 sec
Object Handling	4 min 0 sec	7 min 10 sec	8 min 40 sec	19 min 50 sec
Adult H/F in View	17 min 15 sec	10 min 10 sec	21 min 25 sec	48 min 50 sec
Infant Vocalizations	29 min 25 sec	9 min 0 sec	14 min 40 sec	53 min 5 sec

Table 3. Each infant's age, the amount of video that was coded, time they spent handling objects or watching an adult handle an object, time they spent looking at an adult hand or face in view, and time they spent vocalizing.

	Adult Hand or Face in View	Object Handling	Infant Vocalizations
Total Time	48 min 50 sec	19 min 50 sec	53 min 5 sec
Proportion of Total Time	.35331	.14349	.38406

Table 4. The total time infants spent looking at an adult hand or face, handling an object, or vocalizing. The proportion of each of these events relative to total time is also listed.

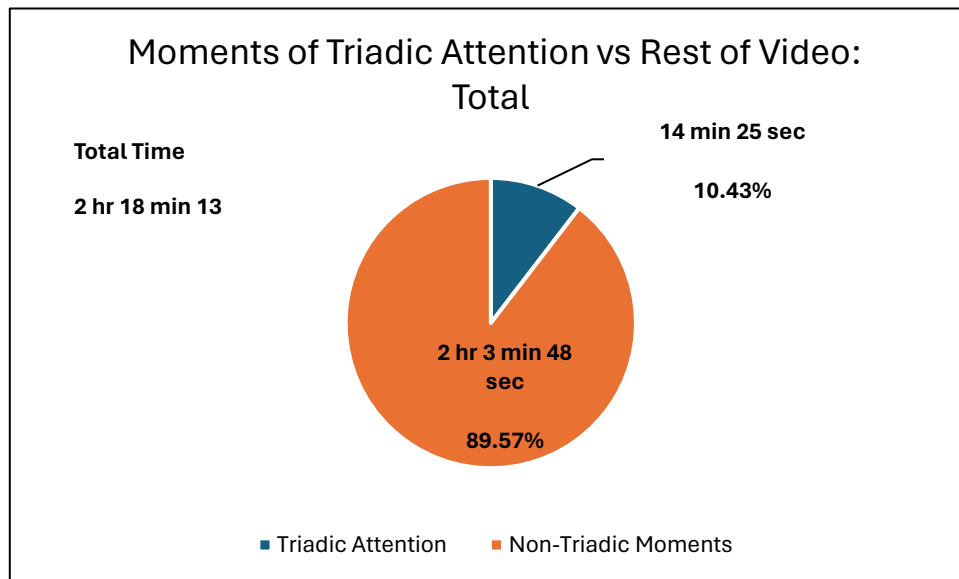


Figure 2. The combined time of all the infants' moments of triadic attention compared to their total time spent outside of triadic attention. Throughout the videos, infants spent most of their time outside of triadic attention.

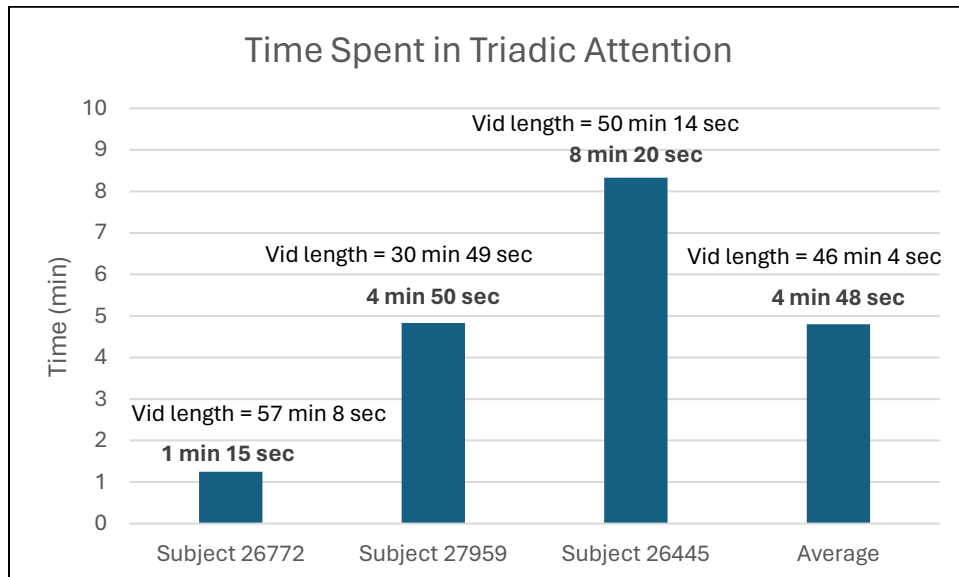


Figure 3. The amount of video each infant contributed, as well as the amount of triadic attention contributed. As infant age increases, time spent in triadic attention also increases.

total time), and Subject 26445 (43 weeks, video length 50 minutes and 14 seconds) produced 8 minutes and 20 seconds of triadic attention (0.166 of total time). These times, as well as the minutes of triadic attention per hour for each subject, are listed in Table 5. From these results, we can conclude that there is a general positive trend of infant age and time spent in triadic attention. Although the video length is inconsistent, there is still a correlation that points in the direction of supported literature (Mundy et al., 2007; Adamson & Bakeman, 1985).

Infant Vocalizations Prior to Triadic Attention

Looking at Figure 4, infants spent 14 minutes and 25 seconds in moments of triadic attention across all videos. For this graph, the amount of time infants spent in triadic attention was tracked by counting the five-second bins. After those were counted, that number was split by whether that triadic attention bin was preceded by a five-second bin containing an infant vocalization. Both the infant vocalization group and the non-infant-vocalization group times were divided by the total triadic attention time to get the proportion listed in Figure 4. From these methods, it was found that infant vocalization occurred immediately prior to a triadic attention event for 4 minutes and 15 seconds, or 29.479% of the overall triadic attention time. Within that same 14 minute and 25 second span, a triadic attention bin without an infant vocalization in the bin immediately prior occurred for 10 minutes and 10 seconds, or 70.521% of the time. These results help draw the conclusion that the presence of an infant vocalization is not necessary to propel an infant into a triadic attention event.

Chi-Square Test for Association

A chi-square test for association, as well as a contingency table, was implemented for each subject, as well as for the total of all subjects. For the total videos, there were 51 bins (4 minutes and 15 seconds) that contained an infant vocalization immediately prior to a moment of triadic

	Subject 2772	Subject 27959	Subject 26445	Average
Total Time	57 min 8 sec	30 min 49 sec	50 min 14 sec	46 min 4 sec
Proportion of Triadic Attention Relative to Total Time	0.022	0.157	0.166	0.104
Triadic Attention/Hour	1.313 min of TA/hr	9.410 min of TA/hr	9.954 min of TA/hr	6.252 min of TA/hr

Table 5. Each subject's total video time contributed, proportion of triadic attention relative to total time, and the number of minutes spent in triadic attention per hour based on that previous proportion.

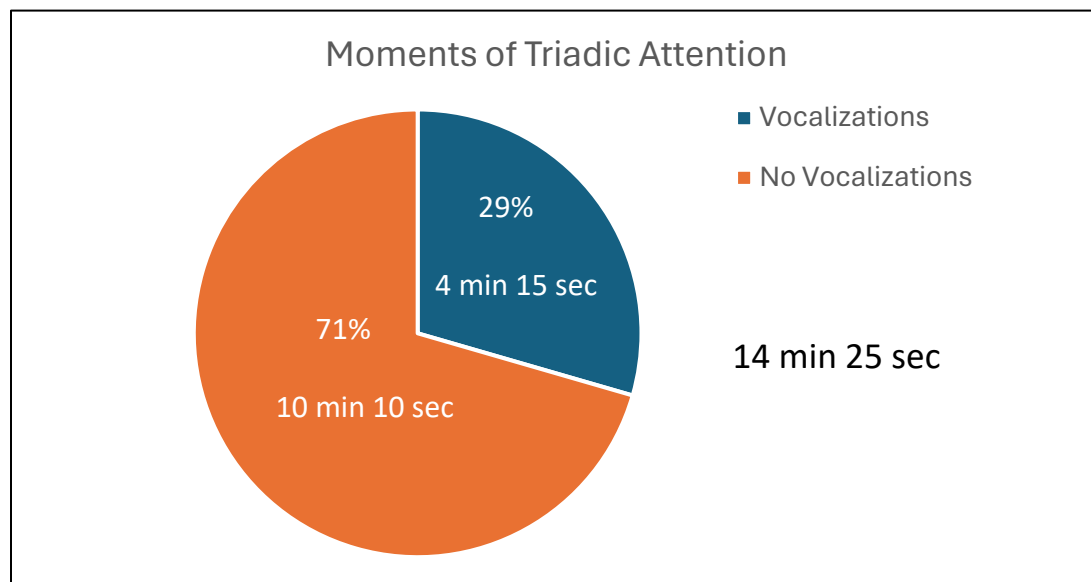


Figure 4. The percentage of triadic attention moments with an infant vocalization directly preceding them compared to the triadic attention moments with no infant vocalizations directly in front.

attention. There were 586 bins (48 minutes and 50 seconds) where an infant vocalization occurred without a moment of triadic attention following it. There were 122 bins (10 minutes and 10 seconds) where a moment of triadic attention occurred without an infant vocalization preceding it. There were 909 bins (1 hour, 15 minutes, and 45 seconds) where a moment of triadic attention did not occur, and an infant was not vocalizing immediately prior to it. The results of the test show that the presence of an infant vocalization event is statistically related to the presence of a triadic attention event. Specifically, infant vocalizations were associated with being more likely to occur before a nontriadic attention event. A total chi-square test accounting for all subjects can be found in Figure 5, and individual subject chi-square tests are listed in Figures 6, 7, and 8 below.

In total, 173 bins (14 minutes and 25 seconds) were moments of triadic attention, 1,495 bins (2 hours, 4 minutes, and 35 seconds) were bins not containing a triadic attention moment, 637 bins (53 minutes and 5 seconds) contained an infant vocalization, and 1,031 bins (1 hour, 25 minutes, and 55 seconds) did not contain an infant vocalization.

The chi-square test returned a value of $X^2 = 6.20$, $N = 1,668$, $df = 1$, $p = 0.013$, which was statistically significant.

Infant Vocalizations Prior to Aspects of Triadic Attention vs. Triadic Attention

Across all videos, 11 minutes and 15 seconds contained an infant vocalization occurring in the bin immediately prior to a bin with an adult face in view. When looking at moments where an infant vocalization occurred immediately prior to a bin with both an adult face and adult hand in view, this occurred for 3 minutes and 20 seconds. To compare, across all videos, infants vocalized prior to a moment of triadic attention in 4 minutes and 15 seconds of the videos.

Total

Contingency Tables			
Contingency Tables			
A	B		Total
	Triadic	Nontriadic	
Number Vocalization Events	51	586	637
Number Non-Vocalization Events	122	909	1031
Total	173	1495	1668

χ^2 Tests			
	Value	df	p
χ^2	6.20	1	0.013
N	1668		

Figure 5. The number of times each infant vocalization category (vocalization or no vocalization) co-occurred with each triadic attention category (triadic attention moment or no triadic attention moment). It also listed the total number of times each subcategory occurred by itself. In the second half of the figure, a chi-square test for association is listed. It states the chi-square value, degrees of freedom, p-value, and total number of occurrences for all four sub-categories combined.

Contingency Tables

Contingency Tables

A	B		Total
	Triadic	Nontriadic	
Number Vocalization Events	9	344	353
Number Nonvocalization Event	6	332	338
Total	15	676	691

 χ^2 Tests

	Value	df	p
χ^2	0.488	1	0.485
N	691		

Figure 6. The number of times -for each individual subject - an infant vocalization category (vocalization or no vocalization) co-occurred with each triadic attention category (triadic attention moment or no triadic attention moment). It also listed the total number of times each subcategory occurred by itself. In the second half of the figure, a chi-square test for association is listed. It states the chi-square value, degrees of freedom, p-value, and total number of occurrences for all four sub-categories combined. The results of the test show that for all three of the subjects, there was no relation between the presence of infant vocalizations and the presence of triadic attention moments.

Contingency Tables			
A	B		Total
	Triadic	Nontriadic	
Number Vocalization Events	20	88	108
Number Non-Vocalization Events	38	223	261
Total	58	311	369

χ^2 Tests			
	Value	df	p
χ^2	0.904	1	0.342
N	369		

Figure 7. The number of times -for each individual subject - an infant vocalization category (vocalization or no vocalization) co-occurred with each triadic attention category (triadic attention moment or no triadic attention moment). It also listed the total number of times each subcategory occurred by itself. In the second half of the figure, a chi-square test of association is listed. It states the chi-square value, degrees of freedom, p-value, and total number of occurrences for all four sub-categories combined. The results of the test show that for all three of the subjects, there was no relation between the presence of infant vocalizations and the presence of triadic attention moments.

Contingency Tables

Contingency Tables

A	B		Total
	Triadic	Nontriadic	
Number Vocalization Events	22	154	176
Number Non-vocalization Events	78	354	432
Total	100	508	608

 χ^2 Tests

	Value	df	p
χ^2	2.81	1	0.094
N	608		

Figure 8. The number of times -for each individual subject - an infant vocalization category (vocalization or no vocalization) co-occurred with each triadic attention category (triadic attention moment or no triadic attention moment). It also listed the total number of times each subcategory occurred by itself. In the second half of the figure, a chi-square test of association is listed. It states the chi-square value, degrees of freedom, p-value, and total number of occurrences for all four sub-categories combined. The results of the test show that for all three of the subjects, there was no relation between the presence of infant vocalizations and the presence of triadic attention moments.

Looking at these pieces of data, it is evident that infants were much more likely to vocalize before only a parent face came into frame. These results are listed in Figure 9.

Across all videos, 637 bins – or 53 minutes and 5 seconds – of bins contained an infant vocalization. This means that infant vocalizations before an adult face was in view in the bin after accounted for 21.193% of all vocalizations, infant vocalizations before an adult hand and face were 6.279% of all infant vocalizations, and infant vocalizations immediately before a moment of triadic attention made up 8.006% of all vocalizations. These results imply that most of the vocalizations infants made were not related to any types of attention, and the rate of infant vocalizations even go down before attention events.

Discussion

At home, in the everyday interactions of 7- to 11-month-old infants with a mature social partner, triadic attention is rare and vocalizations by the infant were less common in these shared attentional moments than outside them. Moments of triadic attention with infant vocalizations immediately prior were not more prevalent than moments of triadic attention with no infant vocalizations immediately prior. In fact, nearly $\frac{3}{4}$ of all triadic attention moments were not preceded by an infant vocalization.

There was a positive correlation between infant age and the amount of time spent in triadic attention. The 29-week-old infant was in bouts of triadic attention the least, then came the 39-week-old infant, and the 43-week-old infant spent the most time in triadic attention. Along with these findings, the total amount of combined time that our infants spent in triadic attention was much smaller than the time spent outside of triadic attention. These findings were consistent with the literature, as 29- to 43-week-old infants should not be spending much of their time in joint attention.

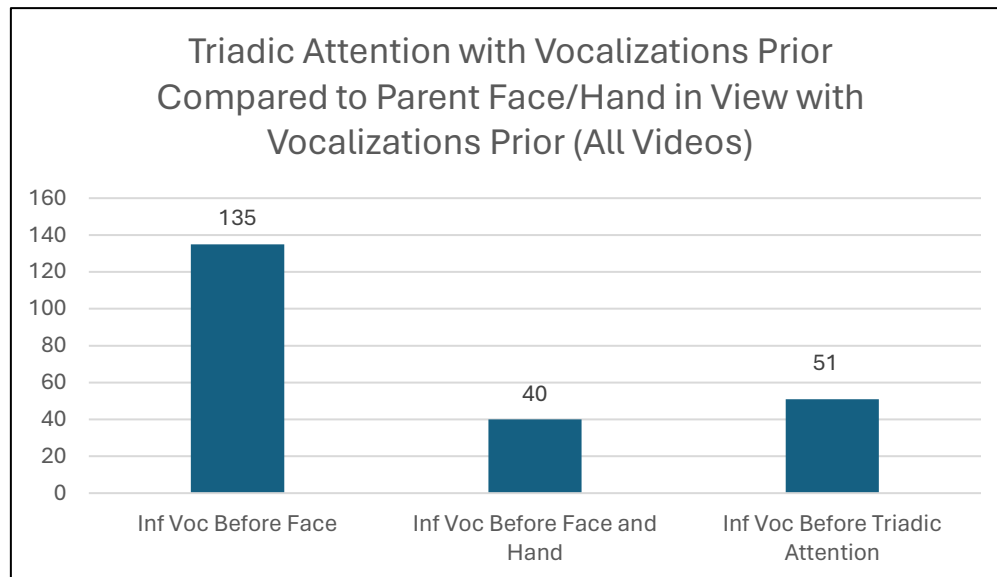


Figure 9. The number of times an infant vocalized in the 5-second bin immediately before a parent face came into frame compared to the number of times an infant vocalized in the 5-second bin immediately before both a parent face and hand came into frame together. These data were also compared to the number of times an infant vocalized before a moment of triadic attention.

Additional Findings

Infant vocalizations tended to occur before face-to-face events without any object handling, moments of shared attention similar to face to face play. Infant vocalizations were more prevalent prior to events where an infant only looked at their parent's face than events where an infant looked at both their parent's face and hand at the same time. Infant vocalizations were also more prevalent prior to events where they only looked at their parent's face compared to infant vocalizations before a triadic attention event. Two possibilities worthy of future research emerge from these observations. First, the high arousal of laboratory visits may elicit more advanced forms of performance (Goldstein et al, 2010). Second, vocalizations during face-to-face interactions may be a precursor to the later role of infant vocalizations in eliciting joint attention to an object.

Infant Vocalizations Did Not Occur Before Triadic Attention Events.

Infant vocalizations did not precede triadic events. There are two further reasons why this may have occurred. First, vocalizations are not the only way infants signal parents of their interest and attention state; object handling and manipulation may be a larger facet compared to vocalizing. Chang et al. (2016) found that between the ages of 4 and 9 months, mothers labeled objects more when their infant handled the object more. Also, there was no correlation between infant vocalizations and maternal object labeling. Since parental object labels are known to initiate moments of triadic attention (Namy et al., 2000), infant pointing may have played a larger role in locking infants into events of triadic attention than I originally thought. Future work needs to track how many object-directed gestures occurred prior to moments of triadic attention compared to infant vocalizations. It would also be beneficial to look at the frequency of

parental object labels and whether parents were labeling objects before or during moments of triadic attention in relation to infant gesturing.

Bakeman and Adamson (1984) discuss how as infants age, they slowly perform less face-to-face interactions with their parents, and more coordinated moments of joint attention involving objects. That early work on joint attention suggested that triadic attention became frequent around the first birthday. Development is incremental –not all or none jumps –and the field now knows there are early beginnings which was the focus of the present study. But is it possible that the infants in this study, particularly in the home context, were too young to show sufficient numbers of triadic events. It would be worthwhile to repeat these analyses with older infants. Additionally, across all videos it was found that infants produced more vocalizations immediately prior to an event where an infant was only looking at their parent’s face compared to immediately prior to a moment of triadic attention. Infants vocalized before face-to-face interactions more than before triadic interactions. This is important information because it tells us that when an infant vocalizes, and at this early age in the development of joint attention (7-11 months) where triadic attention events are not common, face-to-face moments may be the more powerful elicitor of vocal communication. Younger infants may need the face and attention of a mature partner to vocalize (Bakeman & Adamson, 1984).

When Do Objects Play a Larger Role in Organizing Triadic Attention?

Yu and Smith (2013) posit that at the age of one, the object does not just play a large role in triadic attention, it plays the primary role. In their study, they found that infants do not look at parent faces and eyes while in moments of joint attention. Instead, they coordinate attention by looking at an object that they or their parents are handling. This leads to one-year-old infants and parents looking at the same object and forming attention events.

Infants early in development prefer to look at faces compared to looking at objects. At 6 months old, infants will quickly glance first at faces presented to them in the midst of other distractor objects (Gliga et al., 2009). The shift from faces to objects is critical to human social and cognitive development. Clearly more research is needed on how –in perhaps fragile and limited steps – progress is made.

Limitations

The limitations of the study include limited number of infants, since large individual differences in the rate of development of social skills are widely known (Nathani et al., 2006; Bakeman & Adamson, 1984) and the lack of measures of parent behaviors prior to triadic events. An additional limitation is the use of head cameras rather than head -mounted eye-tracking since very brief gaze shifts to the parents could also play a role. Finally, a great deal is not known about infant vocalizations and the contexts outside the lab in which they occur. Here I focused on vocalization in relation to the onset of joint attention, so a more systematic examination of the temporal patterns of vocalization is needed. In particular, infant vocalizations may first operate socially as responses to parent behavior rather than as elicitors.

Future Directions

If I were to replicate this study, I would make two major changes. Firstly, I would track vocalizations during and immediately after triadic attention events. This would allow the current results regarding infant vocalizations to be compared to this new data, and I could examine whether infant vocalizations occurring during or after triadic attention plays a role as to why infants do not vocalize as much before moments of triadic attention. Secondly, I would expand the age range. Ideally, I would use at least three infants from the ages of 3, 6, 9, 12, 15, and 18 months. This way, the entirety of the infant attention spectrum is covered, ranging from initial

face-to-face attention between only the infant and mother, and the much more complex full triadic attention development. Also, infant vocalization types would immensely vary across these ages (Nathani et al., 2006), so I would be able to look at infants with almost no speech-like utterances to infants forming complex syllabic patterns and words. This wide range could play a role in eliminating a compounding variable of age when examining infant vocalization's influence on triadic attention.

To complement the original study, I would like to perform a lab study where mothers are not allowed to initiate moments of triadic attention for a section of the time. I would allow three 30-minute play sessions between each infant and their respective mother, with standard infant toys included in the play area. For one 30-minute play session, I would have the mother and infant play normally, with no further instructions besides those. For the next 30-minute session, I would instruct the mother to focus on initiating moments of triadic attention with her child. Then for the last session, I would instruct the mother to never initiate any moments of triadic attention. Looking at the results from these from play sessions, I will be able to see whether infants are inclined to initiate triadic attention when their parent does not do it. This can lead to an interesting discussion related to the role each party plays in the initiation of triadic attention.

Conclusion

The process of infants vocalizing, as well as engaging in attentional bouts with others and objects that furthers their learning, are essential for cognitive development. What the infant says, how long they are saying it, and why they are saying it are all important aspects of these interactions that need to be understood. This research points towards the idea that infants from the age of 7-11 months do not initiate triadic attention with their voice. Maybe this is true for all infants, but most likely this research needs to be spread to more age groups and specialized

populations. Speech production at these infant ages lays the foundation for later speech production, so identifying any patterns as to when these vocalizations occur is crucial. Children who do not babble at this age (or babble less) are at a higher risk for language difficulties, so identifying moments to locate differences in these speech patterns from infant to infant is crucial. Although it is a small role, this study informed the literature of an age group's vocalization frequency and timing in the naturalistic setting, and how that relates to moments of attention that are crucial to children learning.

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